

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

EEE 304
Power Electronics Laboratory
Final Project Report

Section: C1 Group: 03

Password Based Gate Locking System

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Academic Honesty Statement:

"In signing this statement, we hereby certify that the work on this project is our own and that we have not copied the work of any other students (past or present) and cited all relevant sources while completing this project. We understand that if we fail to honour this agreement, we will each receive a score of ZERO for this project and be subject to failure of this course."

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1. Abstract:

Security is one of the biggest requirement while building structures of importance, such as banks, office buldings, house buldings etc. Normally big structures have security guards at external barriers. But if criminals get past the external security, what then? For this, internal security is also needed especially to protect sensitive things that might create problems if fallen in wrong hands. One of the best internal security design that can protect our valuable things from falling in wrong hands is password based gate locking system. This type of system matches given password with an already existing one. If matches, the door will open and let the person in. If wrong password is pressed, the system will automatically sound an alarm to let everyone know that an intruder is trying to get in.

2. Introduction:

The primary objective of the project is mainly aimed at providing secure access control using a password authentication mechanism. The system is implemented on a breadboard using components such as comparators, demultiplexers, shift registers etc.

The design begins by taking a decimal input from the user, which is then converted into its binary equivalent. The binary data is processed using the implemented logic circuits to authenticate the password and control the locking mechanism. This approach ensures an efficient design while demonstrating the practical use of core digital logic design.

This report details the development process, including the design methodology, circuit implementation, and testing of the password-based gate locking system. It highlights the project's scope, emphasizing the integration of combinational and sequential logic components, and discusses potential challenges and their resolutions.

3. Design:

3.1. Problem Formulation:

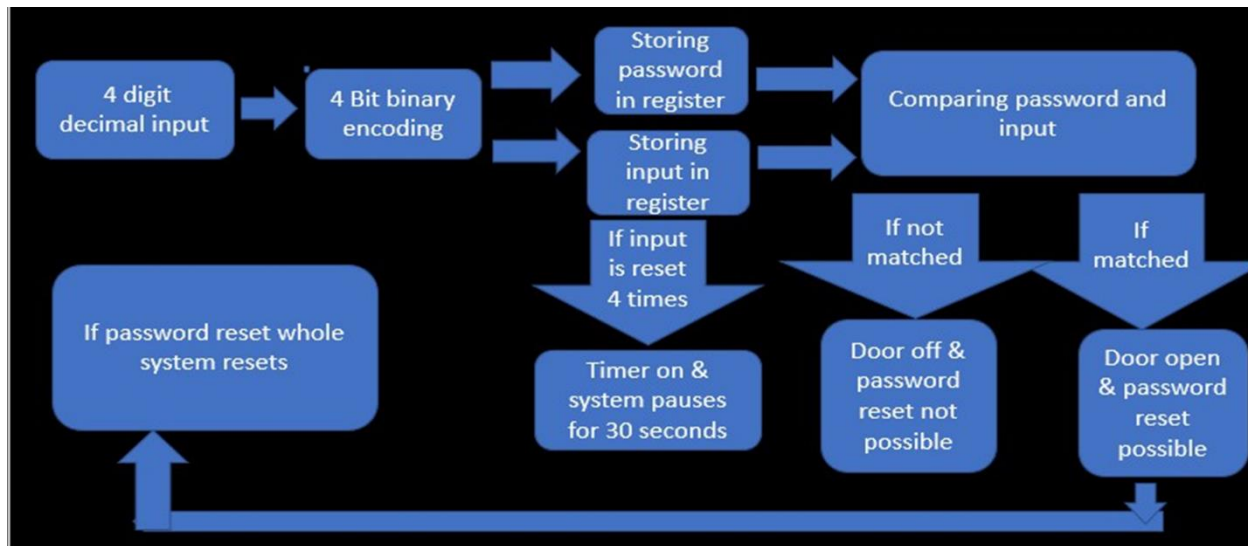
Traditional locking methods, such as mechanical locks and keys, face several limitations, including susceptibility to duplication, physical damage, and misplacement. These issues create vulnerabilities that can lead to unauthorized access and compromised security.

So, we came up with a password-based lock system providing a digital alternative. It takes 4 decimal digits as input and encodes each digit to a 4-bit binary number. So for 4 digits we have a total of 16 binary bits and the same for password input part.

3.1.1. Identification of Scope:

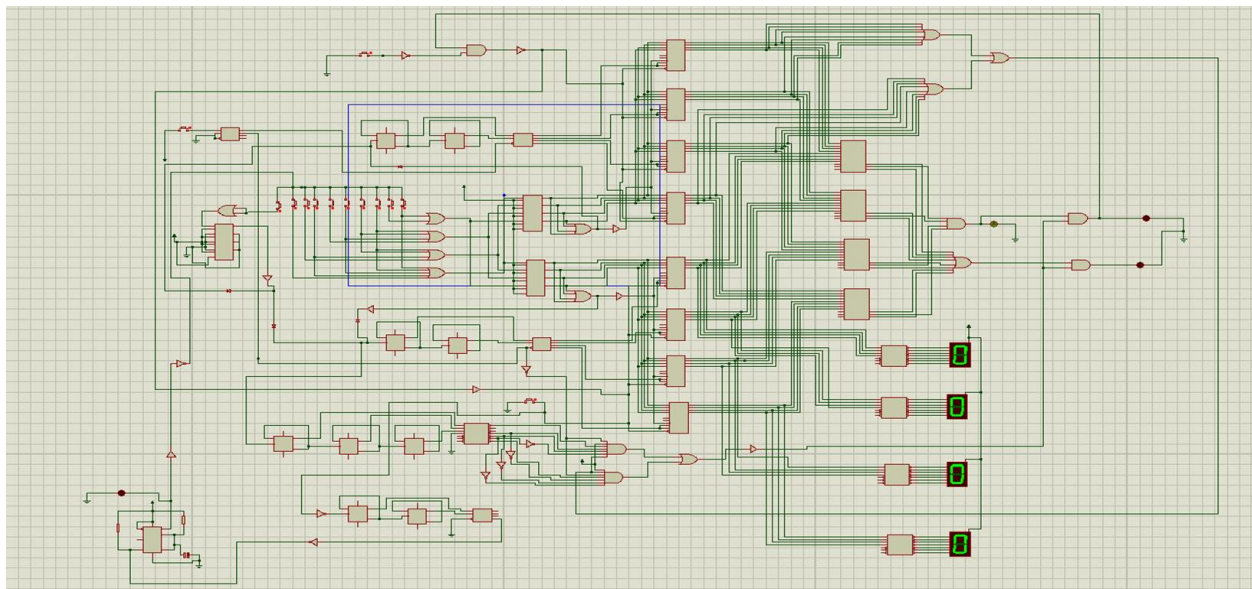
- A detailed study was conducted to identify the appropriate technologies and components required.
- The need for a cost-effective, automated, and secure gate locking solution was identified to address common limitations in traditional locks.
- Analysis was carried out to ensure the system remains cost-efficient.
- The system was investigated for its environmental and operational sustainability.
- The investigation also identified potential risks and challenges like power failure, code vulnerability, false alarms.
- The proposed system fills these gaps by providing a cost-effective, automated, and secure gate locking solution.

3.2 Design Method:

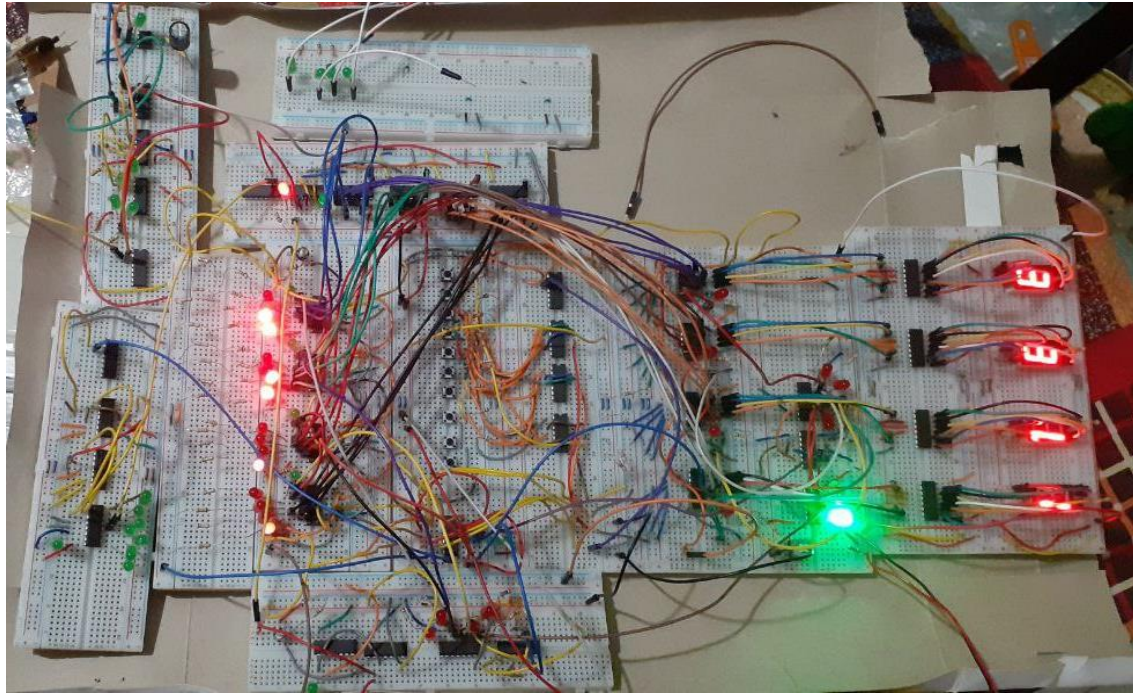


3.3 Hardware Design:

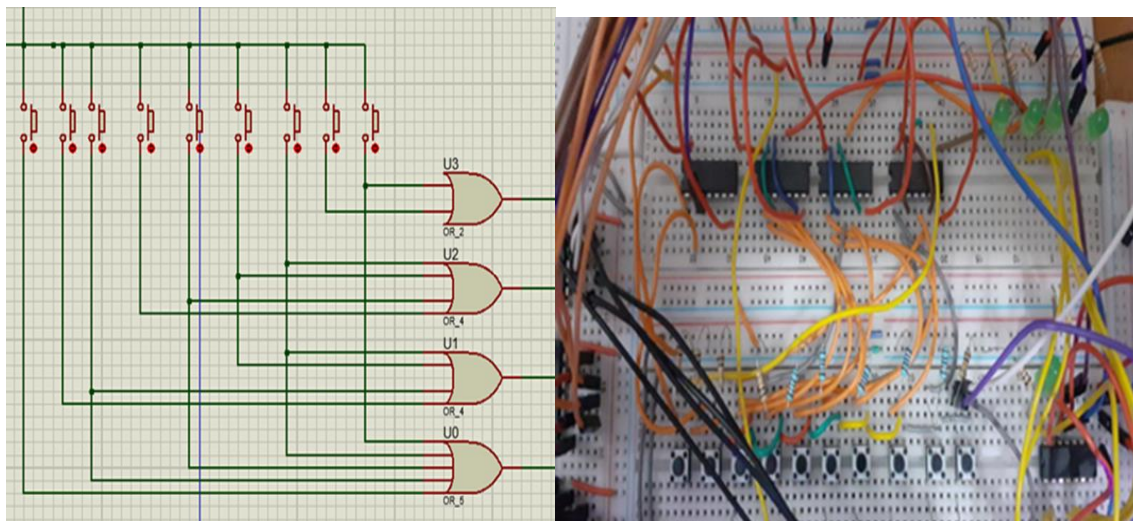
Circuit diagram:



Practical Circuit:

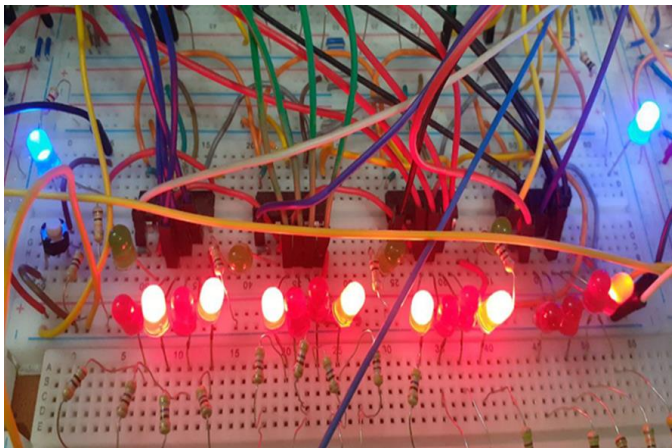
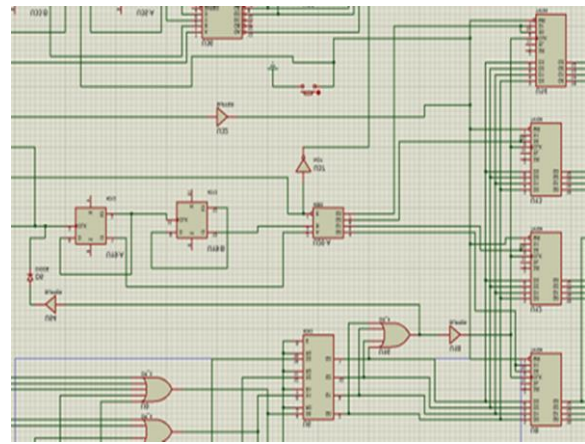
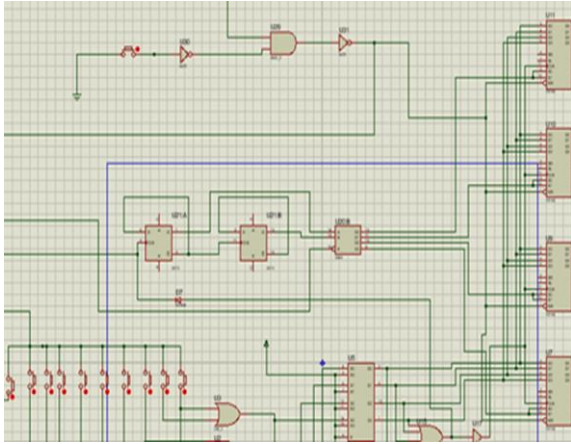


Keypad:



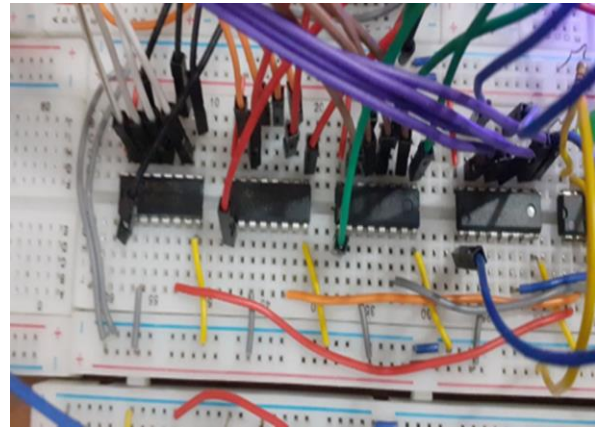
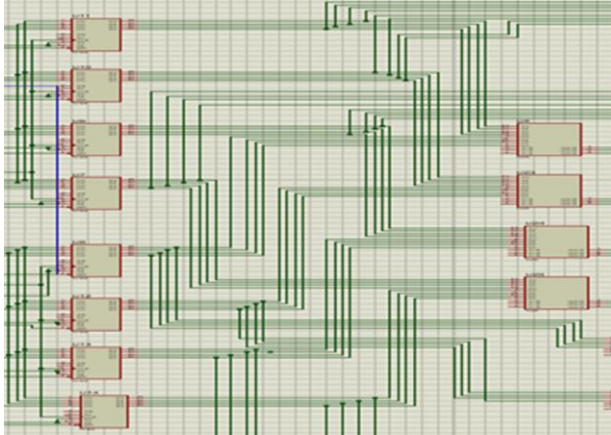
We used 10 push buttons for 0-9 and four or gate to encode.

Shift Register:



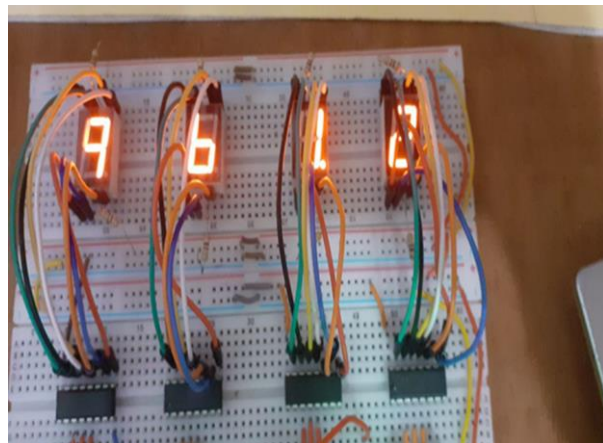
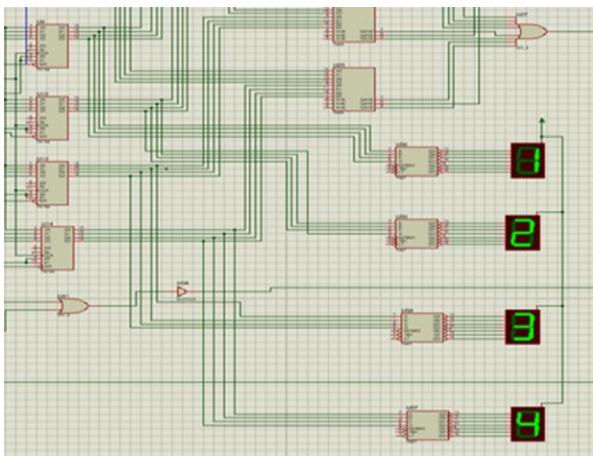
This project involves constructing a shift register to store user passwords and inputs, utilizing 74194 ICs as the main shift registers, 4555 ICs as demultiplexers to direct inputs, and 4013 ICs as D-type flip-flops to manage switching. The 74194 IC stores data with its inputs connected to the 4043 IC, while its clock is triggered by an OR gate. The 4555 IC serves as a switch, automatically routing user inputs to the correct shift register. The 4013 IC functions as a counter, ensuring seamless operation of the demultiplexer when a button is pressed.

Comparator:



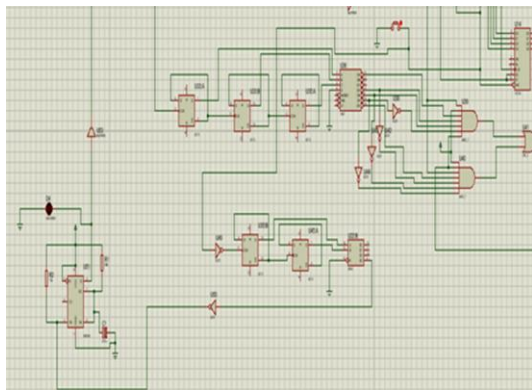
This segment of the project uses eight 7485 ICs to compare the saved password with the user input. The 7485 IC serves as a 4-bit comparator, comparing the values digit by digit based on their place value. A correct password match is determined when $QA = B$ for all bits, while $QA < B$ or $QA > B$ indicates an incorrect input. This ensures accurate authentication of the user's password.

Seven Segment Display:



7447 IC, which decodes binary values into outputs suitable for driving a seven-segment display. The binary inputs A, B, C, and D are connected to Q3, Q2, Q1, and Q0, respectively. The seven-segment display's VCC is connected to a power source, enabling it to visually represent the decoded binary input.

User Input Reset & Delay Timer:

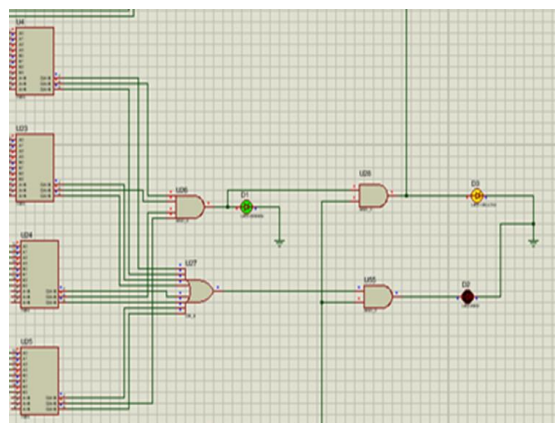


The reset button will allow user to reset input. And a user can reset 4 times successively which is count by a 2 bit counter. After that the delay timer will be on for 30 seconds disabling the system to take any input from the keypad.

The delay time is set using 555 timer IC.

Where $T(\text{delay}) = R \cdot C = 10K \cdot 470\mu F = 30 \text{ sec}$

Final Output:



If all comparator gives the equal pin high and also all 4 digits are inputted only then the final and gate output will be high, and the yellow LED will light up indicating that the password is matched. If password doesn't match, then red led will light up.

4 Implementation:

4.1 Description:

The design of our project was perfectly simulated in Proteus software. Now for the hardware part we needed some basic logic components. The component list is given below:

Component List:

- AND gates, OR gates & NOT gates
- Resistors, Capacitors
- Push Buttons
- Flip flops
- Shift Registers
- Comparator
- Demultiplexer/Decoder
- BCD to 7 Segment Decoder
- Common Anode 7 Segment Display
- LEDs
- Wires & Breadboards

All of these components were used in full effect to complete our project. We proceeded according to our proteus design.

At first, we made the encoder part of our project. For this, we used push buttons and OR gates. We connected them in the logical order that is seen in Encoder ICs. Basically, we manually designed the keypad system that will take 4 decimal numbers as input.

After designing the input system, comes the storing system. We used total 8 Shift Registers to store the decimal numbers. 4 Shift Registers were used to store the input password that will lock the system and 4 other Shift Registers were used to store the input that would be matched with the previously saved password. The outputs of the OR gates went directly to all the 8 Shift Registers. A Flip Flop is also connected to the 2 sets of Shift Registers that selects which Shift register will store value. Also connected with the Flip Flop is a push button that controls the Flip Flop's ability to select the set of Shift Registers.

The comparing system is designed with Comparators. The Comparators used in our project can compare two 4 bit binary numbers . These Comparators are connected to the 2 sets of Shift Register. Comparators are not connected to all Shift Registers. The Comparator which will compare the LSB is connected to the Shift Registers from both sets that store the LSB. The other Comparators are connected in similar way. The Comparators are also connected with two LEDs through OR and NOT gates. The LEDs are there to indicate if the password is correct or incorrect.

The display system consists of BCD to 7 Segment Decoder and Common Anode 7 Segment Display. We have used 4 displays, each for showing a single decimal digit. BCD to 7 Segment Decoders are connected to the two sets of Shift Registers. The 7 Segment Displays are connected to the Decoders. Like Comparators, each Decoder and Display are connected to the specific Shift Register that represents the decimal digit they would decode and display respectively.

Additionally, there are some other ICs, gates and push buttons which provide extra characteristics for our gate locking system such as: Reset option, Protection from multiple try-outs etc.

5. Design Analysis and Evaluation:

5.1 Novelty:

This project combines affordability with innovative features such as real-time code validation, system disabling on incorrect attempts, and alarm activation.

The novelty of the gate locking system project lies in its combination of simplicity, automation, and enhanced security features. Unlike traditional locking mechanisms that rely on physical keys or manual operation, this system introduces a code-based authentication process to control gate access.

The gate locking system project is designed with cost efficiency in mind, making it an affordable yet effective solution for enhancing security.

This combination of low initial investment, low maintenance, and energy efficiency makes the gate locking system a cost-effective solution suitable for homes, small businesses, and other facilities seeking reliable security on a budget.

5.2 Design Considerations:

The design of the project required careful planning and consideration of several critical factors to ensure functionality, reliability, and ease of implementation. The following aspects were taken into account during the design process:

Component Selection: The project utilized digital components such as comparators, demultiplexers, and shift registers to perform logical operations. Components were selected based on their availability, compatibility, and ability to perform the desired functions efficiently.

Input and Conversion: The system was designed to accept a decimal input from the user, requiring reliable conversion of the decimal value to binary. The conversion logic was implemented using standard digital circuit techniques to maintain accuracy and minimize delays.

Password Authentication Logic: The authentication mechanism was built using comparators to verify the user-provided binary password against a predefined binary password. This required precise design of logic circuits to handle binary comparisons effectively.

Signal Control and Data Flow: Demultiplexers and shift registers were used to manage signal control and ensure the correct flow of data within the circuit. The design emphasized minimizing signal interference and maintaining system stability during operation.

5.3 Investigations:

5.3.1 Design of Experiment:

For our experiment, we first reset the whole system so we could enter a password. After that, we entered a 4 digit decimal password and saved it as input password. As, input password was saved, the system was turned on.

Now, came the real experimental part. We entered different variations of decimal inputs as passwords to see how the system responds. Also we used the reset input password option more than once to see the system response.

5.3.2 Results and Analysis:

As we input different variations of password, the system responded accordingly. The system allowed a person to come in only when the right password had been pressed. For the other times, it didn't.

Also, if the input reset button was pressed 4 times, the system would automatically lock itself and won't let the the person to input another password for a certain period of time. So we can say, our project works perfectly.

5.4 Limitations of Tools:

The system relies on a continuous power source to operate. In the event of a power outage, the system may fail unless a backup power source, such as a battery, is integrated.

If the code remains static for extended periods, there is a risk of unauthorized access through guessing, observation, or sharing. Regularly updating the code is necessary to maintain security.

The current system is designed for simple, small-scale applications such as homes or small offices. It may require significant modifications to adapt to larger or more complex security systems.

The basic design does not include remote access or monitoring features, which limits its functionality compared to advanced smart security systems.

5.5 Impact Assessment:

A detailed study was conducted to identify the appropriate technologies and components required.

The need for a cost-effective, automated, and secure gate locking solution was identified to address common limitations in traditional locks.

Analysis was carried out to ensure the system remains cost-efficient.

The system was verified for its environmental and operational sustainability.

The assessment also identified potential risks and challenges like power failure, code vulnerability, false alarms.

The proposed system fills these gaps by providing a cost-effective, automated, and secure gate locking solution.

5.6 Sustainability Evaluation:

The system is designed to consume minimal power, especially when using energy-efficient components like low-power microcontrollers and motors. This reduces the overall energy footprint.

Minimal material usage and the ability to use recyclable components make the project environmentally friendly during its lifecycle.

The system promotes safety and security, which are essential for sustainable living environments. By preventing unauthorized access, it fosters a sense of security for users.

The system can be upgraded with smart features such as remote monitoring, wireless operation, or dynamic code generation, aligning with technological advancements to ensure continued sustainability.

6. Reflection on Individual and Team work:

6.1 Individual Contribution of Each Member:

Project	ID 2006137	ID 2006152	ID 2006154	ID 2006156	Date	Estimated work (in hours)	Estimated duration (in days)
Designing the circuit in proteus	✓	✓	✓	✓	25/8/2024	3	1
Designing the circuit in proteus	✓	✓	✓		27/8/2024	3	1
Designing the circuit in proteus		✓		✓	5/9/2024	3	1
Designing the circuit in proteus		✓	✓		9/9/2024	3	1
Ordering the equipments		✓			12/9/2024	2	1
Receiving the equipments			✓	✓	7/10/2024	5	1
Making Keypad (Encoderr)		✓		✓	8/10/2024	3	1
Shift Resister assembly	✓	✓	✓	✓	11/10/2024	2	1
Shift Resister assembly		✓	✓	✓	14/10/2024	2	1
Comparator assembly	✓	✓		✓	15/10/2024	3	1
Comparator assembly		✓	✓		21/10/2024	1	1
Password reset	✓	✓		✓	29/10/2024	2	1
Delay timer		✓	✓	✓	5/11/2024	2	1
Bcd to 7 seven segmet assembly		✓	✓		15/11/2024	1	1
Final output	✓	✓		✓	26/11/2024	3	1
Final output	✓	✓	✓	✓	30/11/2024	2	1

7 Project Management and Cost Analysis:

7.1 Bill of Materials:

Component Name	Price (Per Piece)(Tk)	Number of Components	Total Price(Tk)
AND , OR, NOT gates	20	8	160
1K, 2K, 5K Resistors	1	30	30
10u, 100u, 470u Capacitors	7	10	70
Push Buttons	6	15	90
Flip Flops	36	5	180
Shift Registers	40	8	320
Comparators	36	5	180

Decoder	46	5	230
BCD to 7 Segment Decoder	46	5	230
Common Anode 7 Segment Display	18	5	90
LEDs	5	40	200
Jumper Wires (Box)	150	4	600
Breadboards	150	15	2250

8 Future Work:

- Remote Access Control: Adding IoT (Internet of Things) capabilities will allow users to control and monitor the gate remotely via a smartphone or computer.
- Mobile App Integration: Developing a dedicated app for code entry, notifications, and system monitoring can enhance user convenience.
- Dynamic Code Generation: Implementing one-time passwords (OTP) or dynamic codes to replace static codes can improve security by preventing code guessing or sharing.
- Biometric Authentication: Adding biometric systems such as fingerprint scanners or facial recognition can enhance security by offering multi-factor authentication alongside the code.

9. References:

- <https://youtu.be/qF75ZS3YjNE?si=JSgnot9l3pb3aR5o>
- <https://youtu.be/bFR4uwCHnAg?si=ItAnW7Ms08v9DPkgE>